

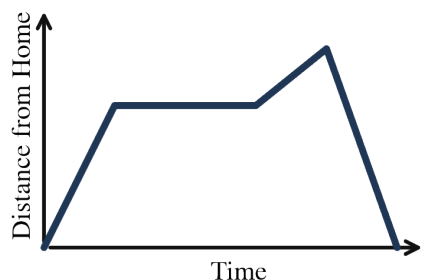
WORKED KEY

Full Name: _____

Period: _____

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

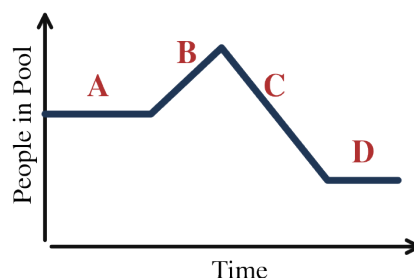
1. Which story matches the graph shown?
Choose only one.



- Ⓐ Zuri rides away from home, turns around right away, and pedals straight back home.
 Ⓑ Zuri walks farther and farther from home the whole time and never stops or turns back.
 Ⓒ Zuri stays home a long while, then walks away and stops at the park for the rest of the day.
 Ⓓ Zuri walks away from home, stays in one spot a long while, moves a little farther away, then heads straight home.
3. Consider the relationship between the number of days Camila waters her tomato plant and the height of the plant. Identify the independent and dependent variables, and explain your choice.

Answer: independent: days of watering;
dependent: plant height

2. The graph shows the number of people at the pool during one afternoon. Match each description to the segment of the graph that fits it best.



- a. Swimmers headed for the exits and the crowd shrank steadily. _____

Answer: C

- b. The pool opened with a small, unchanging group in the water. _____

Answer: A

- c. The last few swimmers kept the count level right up to closing. _____

Answer: D

- d. A rush of swimmers arrived and the count climbed. _____

Answer: B

4. Is the relation $\{(8, 3), (1, 6), (8, 5), (4, 9)\}$ a function? Explain how you know.

Answer: NOT a function — input 8 has outputs 3 and 5

5. State the domain and the range of the relation in problem 4.

Answer: domain: $\{1, 4, 8\}$; range: $\{3, 5, 6, 9\}$

Algebra 1 – Module 3 Practice Test •

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6. Does the table represent a function? Explain.

x	y
4	2
6	5
9	8
6	11

Answer: NOT a function — input 6 appears twice (5 and 11)

7. One vertical line crosses a graph at two points; every other vertical line crosses it just once. Is the graph a function? Explain using the Vertical Line Test.

Answer: not a function — one bad vertical line is enough

8. Let $f(x) = 7x + 4$. Find $f(6)$.

Answer: $f(6) = 42 + 4 = 46$

9. Let $f(x) = 7x + 4$. Find the value of x for which $f(x) = 60$.

Answer: $7x + 4 = 60 \rightarrow 7x = 56 \rightarrow x = 8$

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10. Write a function rule ($y = \dots$) for the table.

x	y
0	2
1	8
2	14
3	20

Answer: $y = 6x + 2$ (starts at 2, climbs by 6)

11. Error Analysis — Gus says a graph that passes through both (2, 5) and (5, 2) must fail the Vertical Line Test. Explain his mistake.

Answer: they sit on DIFFERENT vertical lines ($x = 2$ and $x = 5$) — the test fails only if two points share the same x -value

12. $c(p) = 6p + 25$ models the cost, in dollars, of a party at the trampoline park with p jumpers. Find $c(7)$ and interpret the result.

Answer: $c(7) = 42 + 25 = \$67$ — the 25 is the flat room fee

13. Carnival ride tokens cost \$6 each, and Nia may buy up to 5 tokens. Using tokens as input and total cost as output, state the domain and range of this function.

Answer: domain: $\{0, 1, 2, 3, 4, 5\}$; range: $\{0, 6, 12, 18, 24, 30\}$

Cumulative Review — ACT Style. Circle the best answer.

14. Solve: $3x + 14 = 41$

(A) 5
(B) **9**
(C) 18
(D) 27

Answer: $3x = 27 \rightarrow x = 9$

15. What is the value of $6a - 5b$ when $a = 7$ and $b = 3$?

(E) 12
(F) **27**
(G) 37
(H) 57

Answer: $42 - 15 = 27$

16. Which equation has NO solution?

(A) $7x = 21$
(B) $5(x - 2) = 5x - 10$
(C) $x + 4 = 3x$
(D) **$8x + 3 = 8x - 6$**

Answer: $8x + 3 = 8x - 6 \rightarrow 3 = -6$, false always